

GRAM-SCHMIDT FEATURE SELECTION FOR CLASS ACTIVATION MAPS

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ABSTRACT

Deep Convolutional Neural Networks (CNNs) achieve remarkable performance but lack interpretability, limiting their adoption in high-stakes applications. Existing Class Activation Mapping (CAM) methods suffer from gradient noise sensitivity or computational inefficiency, while lacking principled strategies for selecting informative, non-redundant features. We introduce Gram-Schmidt Feature Selection CAM (GFS-CAM), a novel gradient-free approach that leverages Gram-Schmidt orthogonalization to construct disentangled visual explanations. Our method iteratively builds an orthogonal feature basis by selecting features that explain maximum residual variance. Unlike existing methods that generate single, conflated heatmaps, GFS-CAM isolates distinct semantic concepts into separate, interpretable maps, providing fine-grained attribution of model decisions. Extensive experiments on ILSVRC2012 and PASCAL VOC 2007 across multiple CNN architectures (VGG-16, Inception-v3, ResNeXt-50, DeiT-Base) demonstrate superior performance. GFS-CAM achieves state-of-the-art localization accuracy on three of four architectures, ranks first in 6 of 8 evaluation metrics, and attains best performance on the ROAD faithfulness benchmark for CNNs. Our approach offers a principled solution that combines the robustness of gradient-free methods with computational efficiency, while providing unprecedented interpretability through feature analysis.

Index Terms— Class Activation Mapping, Gram-Schmidt Orthogonalization, Feature Selection, Interpretability, Explainable AI

1. INTRODUCTION

Deep Convolutional Neural Networks (CNNs) have achieved remarkable success across a wide range of computer vision tasks [1]. However, their “black-box” nature presents significant challenges in high-stakes domains such as medical imaging [2], autonomous driving [3], and scientific discovery [4], where interpretability is essential for trust, adoption, and regulatory compliance.

Class Activation Mapping (CAM) techniques address this challenge by highlighting image regions that most strongly influence model decisions. The original CAM required architectural modifications [5], whereas Grad-CAM [6] and Grad-CAM++ [7] leverage gradients to assign importance weights without altering network structure. Despite their effectiveness, gradient-based methods remain sensitive to noise and instability, often producing inconsistent explanations. Gradient-free alternatives have emerged to overcome these limitations. Score-CAM estimates feature map contributions through input perturbation and forward passes [8], while Shap-CAM employs Shapley values to provide theoretically grounded explanations [9]. Although these approaches improve robustness, Score-CAM is computationally expensive, and Shap-CAM often depends

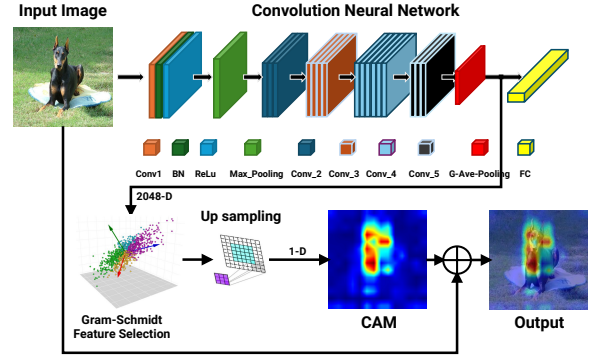


Fig. 1. GFS-CAM schematic: Orthogonalized CNN features using Gram-Schmidt Feature Selection.

on noisy sampling approximations. More critically, existing methods still lack a principled strategy for selecting informative, non-redundant features from high-dimensional activation spaces.

To address these challenges, we introduce *Gram-Schmidt Feature Selection CAM* (GFS-CAM), a novel interpretation method that leverages Gram-Schmidt Feature Selection to identify informative features from the final convolutional layer, as illustrated in Figure 1. Our approach applies Gram-Schmidt orthogonalization to systematically select features that capture core network representations by iteratively removing projections of previously selected features, computing residual variance vectors from the features, and selecting the next feature that maximizes this residual variance, which is then used as the feature weight. This process ensures that the selected features are orthogonal and capture maximal remaining information. GFS-CAM provides a principled gradient-free alternative to existing methods, overcoming the limitations of gradient-based techniques while avoiding the computational overhead of score-based approaches. Moreover, it eliminates nonlinear redundancy and produces disentangled visual explanations. By leveraging Gram-Schmidt orthogonalization, GFS-CAM generates separate, interpretable maps that isolate distinct semantic concepts (e.g., different objects or background regions), offering a more transparent and fine-grained analysis than methods that yield a single, conflated heatmap.

2. RELATED WORK

Class Activation Mapping (CAM) [5] introduced a class-specific visualization by computing a weighted sum of feature maps in the final convolutional layer. Although it highlights the regions most relevant to a predicted class, it requires architectural constraints, global average pooling and a linear classifier, limiting its applicability. Grad-CAM [6] removed these constraints by using the gradients of the

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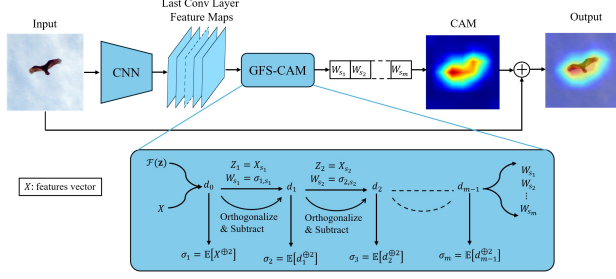


Fig. 2. Schematic of GFS-CAM: The algorithm processes a function family $\mathcal{F}(\mathbf{z})$ and features X from the last convolutional layer. It iteratively substitutes non-random variables z_i in $f(\mathbf{z})$ with Z_i , orthogonalizes the resulting functions, and subtracts their projections from X . This yields random variables $d_j(X)$ and their variance vectors σ_j [10].

target-class score with respect to feature maps to produce importance weights, enabling post-hoc explanations on arbitrary CNNs. Grad-CAM++ [7] further improved localization, especially for multiple object instances, by incorporating higher-order derivatives. Smooth Grad-CAM++ [11] combines Grad-CAM++ with noise-averaging to reduce visual artifacts. LayerCAM [12] aggregates activations from multiple layers to improve spatial detail, and HiResCAM [13] delays downsampling to retain fine features.

To improve robustness, gradient-free alternatives have been proposed. Score-CAM [8] estimates map importance via forward passes on masked inputs, while Ablation-CAM [14] measures confidence drop after removing each feature map. Shap-CAM [9] approximates pixel-wise Shapley values from cooperative game theory at significant computational cost. Dimensionality-reduction approaches such as Eigen-CAM [15] and KPCA-CAM [16] apply PCA and kernel PCA, respectively, to activation maps. By selecting the first principal component as saliency directions, they avoid gradient computations; however, they remain class-agnostic and may highlight non-discriminative features.

3. PROPOSED ALGORITHM

In this section, we explain the proposed algorithm which is based on Gram-Schmidt (GS) Orthogonalization process. Throughout this paper, we make use of the well known Hadamard product, denoted by $\oplus$, where for two vectors $\mathbf{x} = (x_i)_{i=1}^n, \mathbf{y} = (y_i)_{i=1}^n$ we have $\mathbf{x} \oplus \mathbf{y} = (x_i y_i)_{i=1}^n$; similarly $\mathbf{x}^{\oplus 2} = (x_i^2)_{i=1}^n$.

3.1. Gram-Schmidt Orthogonalization over Function Spaces

In this section, we explain the Gram-Schmidt (GS) orthogonalization process over function spaces, upon which all our proposed algorithms are based. As mentioned earlier, we begin by fixing a function family $\mathcal{F}(\mathbf{z})$ in (non-random) variables $\mathbf{z} = (z_1, \dots, z_d)$, but in our experimental sections we focus on multilinear polynomials.

Our algorithm (Algorithm 2) applies in an iterative manner, where in iteration j we select a new feature $Z_1 = X_{s_1}$, substitute $z_j \leftarrow Z_j$ in the functions of $\mathcal{F}(\mathbf{z}_{[j]})$, and perform GS orthogonalization. To this end, Algorithm 1 operates at each iteration j of either of our algorithms by receiving an already orthogonalized $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$, new functions in $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]})$, and a new random variable Z_j . Then, for each member of $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]})$, the algorithm substitutes $z_i \leftarrow Z_i$ for every $i \in [j]$

(Line 6), subtracts its projection on already-orthogonalized functions (Line 7), normalizes (Line 8), and adds the result to $\mathcal{T}$ (Line 9). After the subtraction process we define a new (vector) random variable $d_j(X)$ from X (Line 12), whose variance vector is computed (Line 13) and returned as output.

Algorithm 1 $\text{GS}(\hat{\mathcal{F}}(Z_{\ell_1}, \dots, Z_{\ell_{j-1}}), \mathcal{F}(\mathbf{z}_{[j]})) \setminus \mathcal{F}(\mathbf{z}_{[j-1]}), \{Z_{\ell_i}\}_{i=1}^j$

1: **Input:**

- $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$: an orthogonalized function family in random variables $Z_1, \dots, Z_{j-1}$.
- $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]})$: all the functions in $\mathcal{F}(\mathbf{z})$ which depend on $\mathbf{z}_{[j]}$ but not only on $\mathbf{z}_{[j-1]}$.
- $\{Z_i\}_{i=1}^j$: the random variables $Z_1, \dots, Z_{j-1}$ which appear in $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$, and a new Z_j .

2: **Output:** Orthogonalized function family $\hat{\mathcal{F}}(Z_1, \dots, Z_j)$, and a variance vector σ_{j+1} .

3: **Initialize:** $\mathcal{T} \leftarrow \hat{\mathcal{F}}(Z_{\ell_1}, \dots, Z_{\ell_{j-1}})$.

4: Denote $\mathcal{F}(\mathbf{z}_{[j]}) \setminus \mathcal{F}(\mathbf{z}_{[j-1]}) \triangleq \{f_1(\mathbf{z}), \dots, f_{\ell}(\mathbf{z})\}$, where $f_1(\mathbf{z}) = z_j$.

5: **for** $k \leftarrow 1$ **to** ℓ **do**

6: $g(Z_1, \dots, Z_j) \leftarrow f_k(Z_1, \dots, Z_j)$ (create g from f_k by substituting $z_i \leftarrow Z_i$ for $i \in [j]$.)

7: $g(Z_1, \dots, Z_j) \leftarrow g(Z_1, \dots, Z_j) - \sum_{f \in \mathcal{T}} \mathbb{E}[gf] f$ (orthogonalize g with respect to $\mathcal{T}$.)

8: $g(Z_1, \dots, Z_j) \leftarrow \frac{g(Z_1, \dots, Z_j)}{\sqrt{\mathbb{E}[g(Z_1, \dots, Z_j)^2]}}$ (normalize g .)

9: $\mathcal{T} \leftarrow \mathcal{T} \cup \{g(Z_1, \dots, Z_j)\}$ (add g to $\mathcal{T}$)

10: **end for**

11: Define $\hat{\mathcal{F}}(Z_1, \dots, Z_j) = \mathcal{T}$.

12: Define $d_j(X) = X - \sum_{f \in \mathcal{T}} \mathbb{E}[Xf] f$.

13: Define $\sigma_{j+1} = \mathbb{E}[d_j(X)^{\oplus 2}]$.

14: Return $\sigma_{j+1}, \mathcal{T}$.

3.2. Gram-Schmidt Feature Selection for Class Activation Maps (GFS-CAM)

In Algorithm 2, we propose *Gram-Schmidt Feature Selection CAM* (GFS-CAM), which is inspired by our recent work on Gram-Schmidt methods for feature extraction and selection [17, 18, 10]. At each step j , the algorithm computes the variance vector σ_j to identify the feature of $d_j - 1$ with the highest variance. If the variance of this feature falls below a threshold ϵ^2 , the algorithm terminates; otherwise, it continues by selecting z_j as the highest-variance feature of d_{j-1} , enabling the subsequent call to “GS.” The corresponding variance is then used as the weight for generating the saliency maps.

Generating the GFS-CAM Saliency Map Once the GFS algorithm (Algorithm 2) terminates, it outputs a set of selected feature indices, $\{s_j\}_{j=1}^m$, and their corresponding weights, $\{W_{s_j}\}_{j=1}^m$. These are used to construct the final GFS-CAM saliency map, $L_{\text{GFS-CAM}}$, as a weighted linear combination of the corresponding activation maps from the final convolutional layer. The map is formally defined as:

$$L_{\text{GFS-CAM}} = \text{ReLU} \left(\sum_{j=1}^m W_{s_j} \cdot A^{s_j} \right) \quad (1)$$

where A^{s_j} is the s_j -th activation map. The ReLU function is applied to filter for features that have a positive influence on the explanation.

Algorithm 2 GFS-CAM

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1: Input: A function family  $\mathcal{F}(\mathbf{z})$  and a threshold  $\epsilon > 0$ .
2: Output: Selected features  $s_1, \dots, s_m$  and their corresponding
   residual variances  $W_{s_1}, \dots, W_{s_m}$ .
3: Initialize:  $\sigma_1 = \mathbb{E}[X^{\otimes 2}]$ .
4: for  $j \leftarrow 1$  to  $d$  do
5:   Let  $s_j \triangleq \arg \max_{i \in [d]} \{\sigma_{j,i}\}_{i=1}^d$ , where  $\sigma_j = (\sigma_{j,i})_{i=1}^d$ .
6:   Set  $W_{s_j} = \sigma_{j,s_j}$ .
7:   if  $\|\sigma_j\|_\infty \leq \epsilon^2$  then
8:     break.
9:   else
10:    Define  $Z_j = X_{s_j}$ .
11:   end if
12:    $\sigma_{j+1}, \hat{\mathcal{F}}(Z_1, \dots, Z_j) = \text{GS}(\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1}), \mathcal{F}(\mathbf{z}_{[j]} \setminus$ 
      $\mathcal{F}(\mathbf{z}_{[j-1]}), \{\mathbf{z}_i\}_{i=1}^j)$ 
13: end for

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The resulting raw saliency map is then upsampled to the input image size and normalized to the range $[0, 1]$ for visualization, providing a heatmap that highlights the most informative and non-redundant regions for the model’s decision.

4. EXPERIMENTS

To demonstrate the effectiveness of our proposed method, we conduct a series of experiments to evaluate GFS-CAM against state-of-the-art CAM techniques. The evaluation is performed on standard benchmark datasets and deep learning architectures, using both qualitative and quantitative metrics to assess the quality of the generated visual explanations. We evaluate our method on the ILSVRC2012 validation set and the PASCAL VOC 2007 train-val split using three architectures: VGG-16, Inception-v3, and ResNeXt-50. All input images are resized to 224×224 and normalized using standard ImageNet statistics. For fair comparison, saliency maps are generated from the final convolutional layer of each model and resized to 224×224 .

4.1. Visual Comparison

In our qualitative analysis, we employ ResNet-50 as the backbone network and generate visual explanations using its final convolutional layer. As depicted in Figure 3, GFS-CAM produces the most comprehensive activation maps in comparison to other methods. Grad-CAM struggles to identify the main target objects, often highlighting irrelevant regions, while both Score-CAM and Shape-CAM correctly localize the objects. However, our method exceeds these approaches in accurately pinpointing the relevant regions.

4.2. Evaluation Metrics and Results

To assess the explanation maps, we use standard metrics for faithfulness and interpretability [19, 7, 20]. Faithfulness is evaluated using perturbation tests on the top 20% of salient pixels. The metrics are defined as follows:

Average Drop (AD) [7] measures the confidence drop when only salient regions are shown to the model, defined as $\text{AD} = \frac{1}{N} \sum_{i=1}^N \frac{\max(0, y_i^c - o_i^c)}{y_i^c}$.

Increase in Confidence (IC) [7] is the percentage of images where confidence increases given only the salient region, formulated as $\text{IC} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i^c < o_i^c)$.

Complexity (Com) [19] is the normalized L_1 norm of the heatmap, measuring sparsity, expressed as $\text{Complexity}(x) = \|H_c(x)\|_1$.

Coherency (Coh) [19] measures the stability of an explanation via Pearson correlation, formulated as

$$\text{Coh}(x) = \frac{1}{2} \frac{\text{Cov}(H_c(x \otimes H_c(x)), H_c(x))}{\sigma_{H_c(x \otimes H_c(x))} \sigma_{H_c(x)}} + \frac{1}{2}.$$

As shown in Table 1, our proposed GFS-CAM method demonstrates superior performance across ResNeXt-50 and Inception-v3 on both the ILSVRC2012 and PASCAL VOC 2007 datasets. GFS-CAM ranks first in 6 of the 8 evaluation categories, particularly excelling in Coherency and Complexity, where it secures the top score in 3 out of 4 cases for each. For example, with ResNeXt-50 on ILSVRC2012, it achieves the highest Coherency (69.9) and Complexity (57.4). While other methods like Score-CAM are competitive in certain metrics, GFS-CAM offers the most consistent improvement across the evaluation spectrum, highlighting its effectiveness in generating high-quality visual explanations.

Table 1. CAM comparison on six CNNs and two datasets.

Method	ResNeXt-50				Inception-v3			
	AD ↓	Coh ↑	Com ↑	IC ↑	AD ↓	Coh ↑	Com ↑	IC ↑
Grad-CAM	23.5	38.9	27.7	38.5	26.8	44.6	21.4	45.9
Grad-CAM++	21.3	49.7	24.1	46.1	25.4	40.2	23.6	45.5
Ablation-CAM	26.4	43.3	29.8	44.9	21.5	64.3	29.5	37.1
Score-CAM	16.4	61.8	30.2	53.1	19.4	61.4	28.3	54.7
Shap-CAM	22.1	53.2	37.9	47.2	18.0	52.6	36.4	47.3
GFS-CAM	20.6	69.9	39.8	57.4	21.9	56.4	38.9	58.6
<i>ILSVRC2012 Dataset</i>								
Method	ResNeXt-50				Inception-v3			
	AD ↓	Coh ↑	Com ↑	IC ↑	AD ↓	Coh ↑	Com ↑	IC ↑
Grad-CAM	23.5	42.0	21.4	32.0	26.6	40.2	24.1	30.2
Grad-CAM++	27.1	40.2	30.7	38.3	26.6	38.8	29.0	44.3
Ablation-CAM	21.6	43.3	26.3	51.3	20.0	51.5	25.3	43.2
Score-CAM	20.9	53.7	38.0	47.5	18.7	64.1	30.0	45.1
Shap-CAM	16.5	52.4	30.5	44.2	23.1	54.8	31.3	56.9
GFS-CAM	16.1	58.9	40.9	49.0	22.4	67.7	40.4	60.8
<i>PASCAL VOC 2007 Dataset</i>								

4.3. Localization Accuracy

We evaluate how accurately saliency maps highlight correct image regions using an energy-based pointing game, which measures the proportion of saliency intensity concentrated within the ground-truth bounding box. The evaluation was performed on 2000 randomly sampled images from the ILSVRC 2012 validation set using pre-trained VGG-16, Inception-v3, ResNeXt-50, and DeiT-Base models. As detailed in Table 2, GFS-CAM consistently outperforms prior methods. It achieves the highest localization scores on three of the four architectures, VGG-16 (61.7%), Inception-v3 (61.5%), and DeiT-Base (60.8%), and remains highly competitive on ResNeXt-50. These results demonstrate our method’s robust ability to focus explanations on relevant object regions across diverse models.

4.4. RemOve And Debias (ROAD) Benchmark:

We validate faithfulness using the ROAD benchmark [21], which measures an explanation’s ability to identify critical predictive features (lower score is better). As shown in Table 3, GFS-CAM achieves state-of-the-art results on multiple CNN architectures, outperforming baselines on VGG-16 (53.44), ResNet-50 (-52.8), and

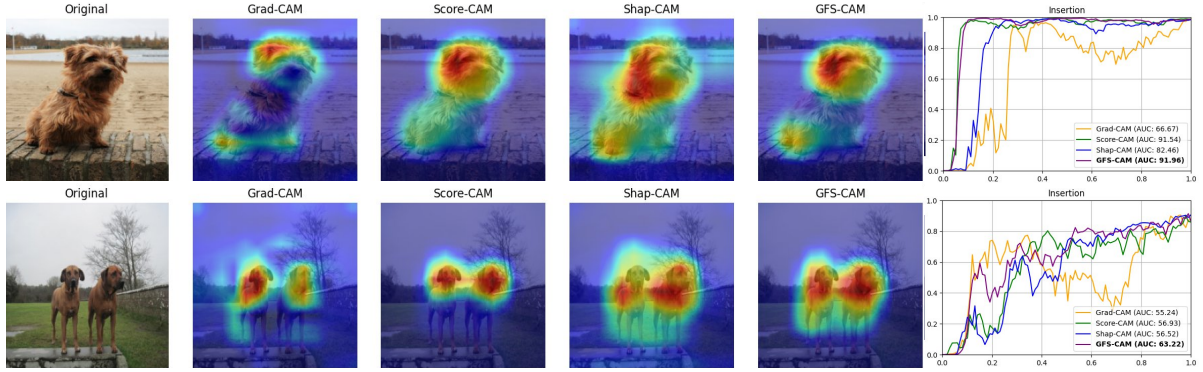


Fig. 3. Grad-CAM, Score-CAM, Shap-CAM, and GFS-CAM generated saliency maps for representative images in terms of insertion curves. In the insertion curve, a better explanation is expected that the prediction score to increase quickly.

Table 2. Pointing game localization accuracy (%) on the ILSVRC 2012 validation set. Higher ($\uparrow$) is better.

Metric	VGG-16	Inception-v3	ResNeXt-50	DeiT-Base
Grad-CAM	46.7	47.6	51.4	51.8
Grad-CAM++	48.5	48.9	53.5	53.2
Ablation-CAM	49.3	48.8	55.6	54.1
Score-CAM	55.5	58.9	57.3	58.4
Shap-CAM	59.2	60.4	59.5	60.1
GFS-CAM	61.7	61.5	59.3	60.8

DenseNet-161 (11.80). While AblationCAM [14] performs best on the DeiT-Base transformer, GFS-CAM’s strong cross-architecture performance highlights its versatility.

Table 3. ROAD benchmark scores ($\times 10^2$), where lower is better ($\downarrow$). GFS-CAM is state-of-the-art on CNNs.

Model	VGG-16	ResNet50	DenseNet-161	DeiT-Base
GradCAM	80.23	-24.3	67.69	5.85
GradCAM++	82.34	-27.8	82.63	9.84
HiResCAM	65.73	3.14	85.40	16.63
ScoreCAM [8]	—	12.1	15.23	10.10
AblationCAM[14]	68.51	-18.3	32.45	3.89
EigenCAM	75.23	-16.2	23.39	11.76
ShapleyCAM	87.57	-24.2	86.42	—
KPCA-CAM	82.03	-58.3	85.32	12.51
GFS-CAM (Ours)	53.44	-52.8	11.80	9.15

4.5. Qualitative Analysis of Feature Separation

GFS-CAM produces disentangled visual explanations by constructing a set of orthogonal features using the Gram-Schmidt process. This allows the decomposition of a model’s prediction into distinct, non-overlapping components. As shown in Figure 4, visualizing the Class Activation Maps (CAMs) for the three highest-variance features reveals that each map localizes a separate semantic concept in the image. This fine-grained attribution contrasts with methods like ScoreCAM and ScoreCAM++ [22], which generate a single, conflated heatmap. GFS-CAM’s approach offers a more transparent tool for understanding the specific visual evidence driving a network’s decision.

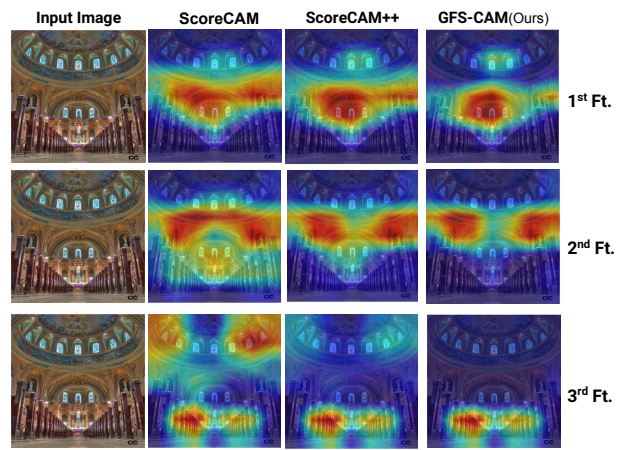


Fig. 4. GFS-CAM generates separate CAMs for orthogonal features to isolate distinct objects, while ScoreCAM and ScoreCAM++ produce a single, conflated map.

5. CONCLUSION AND FUTURE WORK

We introduced GFS-CAM, a principled gradient-free interpretation method that uses Gram-Schmidt orthogonalization to select informative, non-redundant features from CNN activation spaces. Our approach overcomes limitations of existing CAM techniques by providing theoretically grounded feature selection without the computational cost of score-based methods. Experiments across multiple architectures and datasets show that GFS-CAM achieves superior performance, ranking first in 6 of 8 metrics, and produces disentangled visual explanations that isolate distinct semantic concepts into separate, interpretable maps. This offers a more transparent and fine-grained tool for understanding CNN decision-making.

Future work will adapt GFS-CAM to Transformer-based architectures by reformulating the Gram-Schmidt process for attention mechanisms. We also aim to extend the framework to hierarchical and multi-scale explanations through multi-layer processing and enhance computational efficiency with randomized linear algebra techniques.

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